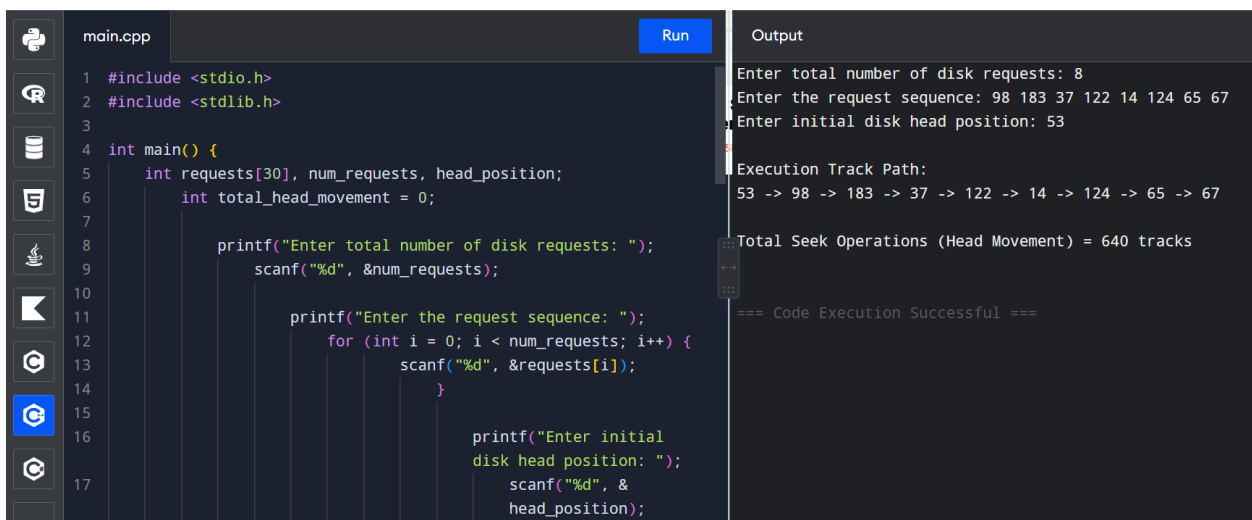


EXPERIMENT 31 :

31. Construct a C program to simulate the First Come First Served disk scheduling algorithm.

AIM:

To construct a C program to simulate the First Come First Served (FCFS) disk scheduling algorithm.



```
main.cpp
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int main() {
5     int requests[30], num_requests, head_position;
6     int total_head_movement = 0;
7
8     printf("Enter total number of disk requests: ");
9     scanf("%d", &num_requests);
10
11     printf("Enter the request sequence: ");
12     for (int i = 0; i < num_requests; i++) {
13         scanf("%d", &requests[i]);
14     }
15
16     printf("Enter initial disk head position: ");
17     scanf("%d", &head_position);
18
19     // Simulation logic for FCFS disk scheduling
20     // (The actual logic is not visible in the provided code snippet)

Output



```
Enter total number of disk requests: 8
Enter the request sequence: 98 183 37 122 14 124 65 67
Enter initial disk head position: 53

Execution Track Path:
53 -> 98 -> 183 -> 37 -> 122 -> 14 -> 124 -> 65 -> 67

Total Seek Operations (Head Movement) = 640 tracks

=== Code Execution Successful ===
```


```

PSEUDO CODE :

BEGIN

PRINT "Enter total number of disk requests:"

READ num_requests

DECLARE requests array of size num_requests

PRINT "Enter disk request queue:"

FOR i FROM 0 TO num_requests - 1 DO

 READ requests[i]

END FOR

PRINT "Enter initial head position:"

READ head_position

```
SET total_head_movement = 0
```

```
SET current_head = head_position
```

```
PRINT "Disk Head Movement Sequence:"
```

```
PRINT current_head
```

```
FOR i FROM 0 TO num_requests - 1 DO
```

```
    SET distance = ABSOLUTE(requests[i] - current_head)
```

```
    SET total_head_movement = total_head_movement + distance
```

```
    SET current_head = requests[i]
```

```
    PRINT " -> " + current_head
```

```
END FOR
```

```
PRINT "Total Head Movement: " + total_head_movement
```

```
END
```

CODE:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int main() {
```

```
    int requests[30],
```

```
    num_requests,
```

```
    head_position;
```

```
    int
```

```
total_head_movement =  
0;
```

```
    printf("Enter total  
number of disk requests:  
");
```

```
    scanf("%d",  
&num_requests);
```

```
    printf("Enter the  
request sequence: ");
```

```
    for (int i = 0; i <  
num_requests; i++) {  
  
        scanf("%d",  
&requests[i]);  
  
    }
```

```
    printf("Enter initial  
disk head position: ");
```

```
    scanf("%d",
```

```
&head_position);
```

```
    int current_head =  
    head_position;
```

```
    printf("\nExecution  
Track Path:\n%d",  
    current_head);
```

```
    for (int i = 0; i <  
    num_requests; i++) {
```

```
        total_head_movement  
        += abs(requests[i] -  
        current_head);
```

```
        current_head =  
        requests[i];
```

```
        printf(" -> %d",  
        current_head);
```

```
    }
```

```
    printf("\n\nTotal Seek  
Operations (Head  
Movement) = %d  
tracks\n",  
total_head_movement);  
  
    return 0;  
  
}
```